

Hackathon Report — Breast Cancer Prediction

Overview

This project was developed as part of a machine learning hackathon focused on breast cancer data analysis and prediction.

The project includes several machine learning tasks:

1. Data preprocessing
2. Predicting metastases across anatomical sites
3. Predicting tumor size
4. Performing unsupervised data analysis to identify meaningful patient subgroups

The original dataset contains private clinical information and is therefore not included in this repository.

Part 0: Data Preprocessing

The first step of the project was to clean and preprocess the raw clinical data.

We began by normalizing missing-value tokens such as "NA" and "-" into standard missing values. Then, we systematically parsed and converted the raw columns into usable formats.

The preprocessing stage included:

- Splitting composite identifiers
- Extracting numeric grades and percentages from free-text fields
- Mapping Hebrew medical terms into English categories
- Converting date-time fields into proper date and time formats
- Processing clinical markers such as HER2 and Ki-67 into numeric scores and categorical indicators
- One-hot encoding categorical variables such as surgery types and TNM tumor categories
- Converting count fields into integer values
- Removing the original raw columns after extracting the relevant information

Finally, the preprocessing was wrapped in a pipeline that imputes remaining missing values and scales continuous features. This produced a clean, fully numeric dataset suitable for machine learning models.

Part 1: Predicting Metastases

Goal

The goal of this part was to predict the presence of metastases across 11 anatomical sites using patient clinical features.

Since a single patient may have metastases in more than one anatomical site, this task was framed as a multi-label classification problem.

Baseline Model: Per-Label Decision Trees

Model Description

As a baseline, we trained a separate decision tree classifier for each metastasis site using `MultiOutputClassifier`.

Each label was treated independently. The model outputs a probability score vector, where each entry represents the predicted probability of metastasis for one anatomical site.

Training Procedure

The baseline training process included:

- Using input features from the cleaned dataset
- Sampling 15% of the training data
- Splitting the sampled data into training and validation sets
- Imputing missing values using an `ImputerPipeline`
- Training a decision tree of fixed depth for each label
- Converting probabilities into binary predictions using a threshold of 0.5

Evaluation

The model was evaluated on the validation set using the provided scoring script.

The evaluation metrics were:

- Micro-F1 score
- Macro-F1 score

Baseline Results

Tree Depth	Micro-F1	Macro-F1
3	0.330	0.090
5	0.424	0.205
10	0.640	0.380
15	0.720	0.420
20	0.737	0.430

The baseline results show that increasing the tree depth improved performance, with the best baseline result achieved at depth 20.

Advanced Model: Neural Network with Decision Tree Scores

Model Description

To improve prediction performance, we introduced a neural network model that uses the decision tree outputs as additional features.

The model works as follows:

1. A separate decision tree is trained for each metastasis label.
2. Instead of converting the tree outputs into binary predictions, we keep the raw predicted probabilities.
3. These probability scores are concatenated with the original input features.
4. The combined feature vector is passed into a feedforward neural network called `BoostedMetastasisNN`.
5. The neural network performs the final multi-label classification.

This approach allows the neural network to learn more complex relationships between the original clinical features and the site-specific risk signals produced by the decision trees.

Training Procedure

The advanced model training process included:

- Imputing missing values using the same `ImputerPipeline`
- Training decision trees as in the baseline model
- Freezing the decision trees and using them to generate probability score vectors
- Concatenating the score vectors with the original input features
- Training a feedforward neural network on the combined input
- Optimizing the model using Adam and binary cross-entropy loss

Hyperparameter Selection

We performed grid search cross-validation over the following hyperparameters:

- Decision tree depth
- Number of epochs
- Batch size
- Learning rate

For each hyperparameter combination, the full pipeline was trained and evaluated. The configuration with the highest average F1 score across validation folds was selected.

Evaluation

The best-performing model was evaluated on a held-out validation set using the same scoring script as the baseline.

Advanced Model Results

Metric	Score
Micro-F1	0.9783
Macro-F1	0.8603

The advanced model significantly outperformed the decision tree baseline, showing that combining decision-tree probability features with a neural network can substantially improve multi-label metastasis prediction.

Part 2: Predicting Tumor Size

Goal

The goal of this part was to predict tumor size using clinical and demographic patient features.

This task was treated as a regression problem.

Methodology

1. Data Processing

We engineered and cleaned the features before training the models.

The preprocessing included:

- Transforming numerical variables to capture non-linear effects
- One-hot encoding categorical variables
- Imputing missing values
- Selecting the most informative predictors

This helped reduce noise and allowed the models to focus on the strongest predictive signals.

2. Data Split

The dataset was divided into:

- Training set
- Validation set
- Test set

This ensured that model fitting, hyperparameter tuning, and final evaluation were performed on separate data splits, reducing the risk of biased performance estimates.

3. Baseline XGBoost Model

As a baseline, we trained an XGBoost model.

XGBoost was chosen because gradient-boosted trees are effective at capturing complex, non-linear interactions between clinical and demographic variables.

4. Stacking Predictions

After training the XGBoost model, we used its predictions as an additional feature for the neural network.

This stacking approach allowed the neural network to benefit from patterns already learned by the XGBoost model.

5. Neural Network Tuning

We tuned the neural network using grid search and early stopping.

The tuning process focused on:

- Network architecture
- Learning rate
- Regularization strength

The goal was to maximize predictive accuracy while reducing overfitting.

6. Final Model Training

After selecting the best hyperparameters, we retrained the model on the combined training and validation data.

The final model was then evaluated on the held-out test set.

7. Test-Set Predictions

Finally, the full preprocessing and modeling pipeline was applied to the unseen test data in order to generate tumor size predictions.

Results

Model	MSE
Baseline XGBoost	1.3288
Final Model	0.3524

The final model achieved a much lower mean squared error compared to the XGBoost baseline, indicating a significant improvement in tumor size prediction.

Part 3: Unsupervised Data Analysis

Goal

The goal of this part was to explore the dataset using unsupervised learning methods and identify meaningful patterns among patients.

The team worked in two directions:

1. Biological subtype discovery using PCA and K-Means clustering
2. Tumor size analysis by hospital and age group

Direction 1: Biological Subtype Discovery

Objective

The objective was to identify biologically meaningful patient subgroups based on hormone receptor expression and tumor morphology.

Methodology

We began by loading the data into a pandas DataFrame and standardizing the column names.

Next, we removed administrative and date-related columns that do not directly represent tumor biology or patient characteristics, such as:

- Patient IDs
- Hospital codes
- Surgery timestamps

We then selected seven key biological features:

- ER
- PR
- HER2
- Ki-67
- Tumor width
- Tumor depth

- TNM T-stage

To reduce dimensionality, we applied PCA to the standardized features and retained the first two principal components.

The first two principal components explained approximately 54.0% of the total variance:

Component	Explained Variance
PC1	37.7%
PC2	16.3%

After PCA, we applied K-Means clustering with $k = 3$ to divide patients into three subgroups.

Results

The clustering revealed three main patient groups:

Cluster	Description
Cluster 1	Highest hormone marker levels and smaller tumors. This may represent a hormone-driven subtype with lower tumor burden.
Cluster 2	Lowest receptor levels and larger, more advanced tumors. This may represent a more aggressive, non-hormone-driven phenotype.
Cluster 0	Intermediate hormone expression and intermediate tumor size.

Although no strong unique conclusion was found, the analysis suggests that additional features may help reveal more meaningful characteristics of each patient subgroup.

Direction 2: Tumor Size by Hospital and Age Group

Objective

The objective was to compare tumor size distributions across hospitals and age groups.

The analysis was performed after removing outliers in order to obtain a cleaner and more reliable view of the differences between hospitals.

Findings by Age Group

Young Age Group

Hospital 20 showed especially small tumors. The median tumor size was close to 0.5 mm, and most observations were relatively low.

Hospitals 60 and 62 showed significantly larger tumors, with median values around 1.5–2 mm.

Hospital 85 was in the middle, slightly lower than Hospital 62.

Overall, there was noticeable variation between hospitals in the young age group, even after removing outliers.

Middle-Aged Group

In the middle-aged group, all hospitals showed median tumor sizes between approximately 1 and 1.7 mm.

Hospital 62 remained relatively high, similar to its behavior in the young age group.

However, the differences between hospitals were smaller compared to the young group.

This may indicate that age reduces the variation between hospitals.

Older Age Group

In the older age group, Hospitals 60 and 62 still had slightly higher median tumor sizes.

Hospital 85 consistently showed smaller tumors.

The spread remained wide across all age groups, but without extreme jumps.

Overall, the differences between hospitals appeared smaller among older patients.

Summary of Unsupervised Analysis

After removing outliers, the analysis provided a cleaner picture of the differences between hospitals.

The main observations were:

- Hospital 62 tended to show larger tumors across all age groups.
- Hospital 20 stood out with very small tumors among young patients, but became more similar to the other hospitals as age increased.
- The variation between hospitals decreased as patient age increased.

This trend may be related to more uniform clinical policies among older patients, or to earlier detection among younger patients in certain hospitals.

Final Summary

This project explored several machine learning approaches for breast cancer prediction and analysis.

The main achievements were:

- Building a full preprocessing pipeline for private clinical data
- Developing a multi-label metastasis prediction model
- Improving the baseline metastasis model using decision-tree probability features and a neural network
- Predicting tumor size using XGBoost and neural network stacking
- Performing unsupervised analysis to explore biological subtypes and hospital-level patterns

Overall, the project demonstrated how different machine learning techniques can be applied to clinical breast cancer data, including classification, regression, and unsupervised exploratory analysis.